Considerations for Future Development

This document will outline a few brief pieces of technology that are used for the phase one implementation of Allies Connect. These currently work in a development context and at a small scale, like for the site's use in Georgia, but as Allies Connect expands they may need to make additional considerations for the future.

Google Maps

As the site exists currently, Google Maps is used as the mapping provider for the resource page. To put things simply, this means that every time someone navigates to the maps page the Google Maps API gets called to load the map. For a given Google account the map platform API can be called 10,000 times for free. After 10,000 events, the events begin to incur charges. For the next 90,000 events the cost per 1000 events is $2.83. And for the next 400,000 events after that the cost is $2.27 per 1000 events. For a full month of Maximum usage this would cost around $500. Therefore, the current implementation works well for small-scale development and deployment, but if the site gains too much traffic, this may become too costly to sustain.

Billable events		Cost per 1000 billable events	Calculated cost per pricing tier
0 - 10,000 free cap	First 10,000 events	$0	$0 * 10,000/1,000 = **$0.00**
10,001 - 100,000	Next 90,000 events	$2.83	$2.83 * 90,000/1,000 = **$254.70**
100,001 - 500,000	Next 400,000 events	$2.27	$2.27 * (*200,000* - 100,000)/1,000 = **$227.00**
TOTAL MONTHLY COST			**$481.70**

An effective alternative to Google Maps would be Mapbox. Mapbox provides a map SDK with additional features like navigation and geofencing for free. We did not initially implement this in Phase 1 because mapbox has a slightly more complex implementation process than Google maps. However, it may be worth the development time to switch over to mapbox if the maintainers for Allies Connect wish to scale up the system beyond just Georgia. Another alternative would be OpenStreetMaps. OpenStreetMaps is a community driven open source project where all the functionality is free. However, these community sourced maps may not be as up to date as the Google maps, which is why we did not choose it initially for this project.

Database Management System

For Phase 1 development of Allies Connect our team utilized a service called Free SQL Database as the service to host the relational database for Allies Connect. We chose this service specifically because it was an entirely free way to store data in a relational database, so as to comply with the requirements set out for this project. However, one of the drawbacks to Free SQL Database is that the storage capacity of the provided database is limited.

In the deployment guide for our project, we recommend setting up the website on an Amazon Lightsail instance. If this is the direction that the maintainers for Allies Connect choose to go with, the AWS services offer relational databases for MySQL starting at a small fee. We strongly recommend this, since different databases use slightly different variance of sql. Since Free SQL Database already uses MySQL as the format for the relational database, none of the existing database queries that we developed would need to change. Additionally ,keeping everything on AWS has its own benefits since everything on the Amazon ecosystem is designed to effectively work with itself. Hosting on AWS, as with any cloud provider does have a few downsides. Instead of relying on your own company servers you are relying on the uptime of the cloud provider's servers. Meaning, if AWS were to go down, the hosting service for the site would go down as well.

If AWS is not ideal for hosting the Allies Connect service, there are other cloud database providers as well. Digital Ocean is designed for small-scale deployments, though that may lead to issues down the line as Allies Connect expands its reach. Google provides cloud hosting through the Google cloud and Microsoft provides database hosting through Azure as well. The alternative to cloud-based hosting would be hosting the database on servers owned and operated by the Allies Connect team. This would give the team more control over deployment of new features and the ability to manage the server data more closely, however it is highly likely that they would need to employ an IT professional to keep their servers secure and running consistently.

In the meantime, to help cut down on database storage costs, a component of Phase 2 development could be focused on implementing features that help trim the amount of unused data in the database. For instance, automatically deleting events after a certain amount of time has passed since the event took place, or deleting old shifts after a certain number of months have passed since the shift ended. This would require a tight balance of determining which information is necessary to keep for recordkeeping purposes and which information is okay to discard. That discussion would have to be held between the phase two implementation team and the maintainers of Allies Connect.